

Cross-Dataset Deepfake Detection with a Vision Transformer Mixture-of-Experts Model and SBI-Based Pretraining

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1 Introduction

Deepfake generation methods have made synthetic face manipulation increasingly realistic, raising concerns for media integrity, online misinformation, and the reliability of visual evidence. As a result, deepfake detection has become an important research problem in computer vision and machine learning. The central goal is to distinguish authentic facial content from manipulated content in a way that remains reliable under practical conditions, where the quality, style, and production pipeline of manipulated media may vary substantially.

Despite rapid progress, one of the most difficult challenges in this area is cross-dataset generalization. Many detection models report strong performance when they are trained and evaluated on the same benchmark, yet their accuracy often drops markedly when tested on a different dataset. This issue is particularly visible when a model trained on FaceForensics++ (FF++) is evaluated on Celeb-DF. Although both datasets are used for deepfake detection, they differ in their visual characteristics, manipulation artifacts, compression patterns, and overall data distribution. Consequently, a detector that appears effective in-domain may prove less reliable out-of-domain.

This observation motivates the main focus of this report: deepfake detection should be treated not only as a classification problem, but also as a robustness problem under domain shift. High in-domain accuracy alone is insufficient if the learned representation depends too strongly on dataset-specific shortcuts. In this project, the goal is to study whether cross-dataset robustness can be improved for a Vision Transformer (ViT-B/16) Mixture-of-Experts (MoE) detector by introducing SBI-style synthetic forgery pretraining [?] before supervised training on FF++. The core question is whether such pretraining can encourage the model to learn more transferable manipulation cues prior to adaptation to a specific benchmark.

2 Background and Related Work

2.1 CNN-Based Detectors

Existing deepfake detectors have largely been developed along two complementary lines. The first line is based on convolutional neural networks (CNNs), which are effective at capturing local

image statistics and fine-grained visual artifacts. This makes CNN-based detectors well suited for identifying blending boundaries, texture inconsistencies, and other localized traces of manipulation. However, this strength is also a limitation. Because many of these cues are closely tied to the generation or compression process used in a given dataset, CNN-based models often overfit to dataset-specific artifacts and show limited robustness when evaluated on unseen benchmarks.

2.2 Vision Transformer

The second line of work uses Vision Transformers (ViTs) [2], which build on transformer architectures and self-attention mechanisms [1] to model longer-range relationships and global semantic structure more naturally than standard CNNs. In deepfake detection, this global modeling capacity is attractive because manipulated content may introduce inconsistencies across broader facial regions rather than at a single pixel neighborhood. At the same time, pure transformer-based models may underemphasize subtle local signals that remain useful for forensic analysis, especially when these signals are weak or spatially sparse. As a result, ViT-based detectors can also face difficulty in cross-dataset settings: they may capture useful holistic structure, but still fail to retain the precise local sensitivity needed for robust out-of-domain discrimination.

Taken together, these observations suggest that neither purely local nor purely global modeling is sufficient on its own. Both CNN-based and ViT-based detectors can achieve strong benchmark results, yet both often struggle to generalize across datasets because the learned features remain entangled with the properties of the training distribution.

2.3 MoE-FFD

One recent direction designed to address this modeling tension is the MoE-FFD framework [5]. Its central idea is to combine a frozen ViT backbone with multiple lightweight expert modules that specialize in complementary forms of evidence. In particular, LoRA experts [4] are used to adapt global representations efficiently, while adapter-based experts are introduced to capture more localized forgery cues. A routing mechanism, conceptually related to sparsely-gated Mixture-of-Experts formulations [3], then selects or emphasizes experts dynamically, allowing the model to allocate capacity according to the input. This Mixture-of-Experts design increases flexibility without requiring the entire backbone to be retrained, and it provides a principled way to combine global and local information within a single detector.

However, the existence of a stronger architecture does not eliminate the generalization problem. Even if MoE-FFD improves the representation capacity of the detector, it is still typically trained on a single supervised dataset such as FF++. Under this regime, the model may continue to learn patterns that are highly predictive within that dataset but less informative elsewhere. In other words, MoE-FFD offers an architectural mechanism for modeling richer forgery cues, but it does not by itself guarantee that those cues will be dataset-invariant. This project therefore treats MoE-FFD as a strong baseline and asks whether its cross-dataset behavior can be improved through a better training strategy.

3 Problem Statement

The research problem addressed in this report is the improvement of cross-dataset generalization in deepfake detection. More specifically, the project investigates whether a detector trained primarily on FF++ can maintain stronger performance when evaluated on Celeb-DF, which serves as an out-of-domain benchmark.

This problem remains difficult even for strong models. A MoE-based detector with a ViT backbone, LoRA experts, adapter experts, and routing has considerable representational capacity, but that capacity can still be used to fit dataset-specific regularities rather than general manipulation cues. When this happens, the model performs well on in-domain data yet degrades on data drawn from a different distribution. The central challenge is therefore not merely to improve performance on FF++, but to reduce overfitting to FF++-specific artifacts so that the learned representation transfers more reliably across datasets.

4 Proposed Approach

The proposed approach in this project is to combine the MoE-FFD detector with SBI-style synthetic forgery pretraining [6] before standard supervised training on FF++. This idea is framed as a hypothesis rather than a guaranteed improvement: if the model is first exposed to synthetic blended manipulations derived from authentic images, it may learn a more general notion of forgery inconsistency before adapting to the artifact profile of a specific deepfake dataset.

The intuition is data-centric. Instead of introducing a new backbone or a different expert architecture, the project modifies the training pipeline so that the model first learns from controlled synthetic forgeries and only afterward specializes to the real deepfake distribution in FF++. SBI-style pretraining is a reasonable candidate for this role because it can create manipulated samples with explicit blending-based inconsistencies while remaining closely tied to real facial content. This may encourage the detector to focus on transferable irregularities, such as mismatched boundaries, semantic inconsistency, or unnatural compositional transitions, rather than depending too heavily on the narrow artifacts of a single benchmark.

Accordingly, this project should be understood as a data-centric extension to the MoE-FFD baseline. The architecture remains the same in the main comparison; what changes is the sequence of supervision used to shape the representation. The main hypothesis is that staged pretraining with SBI-style synthetic forgeries can improve out-of-domain robustness, even if the resulting model does not maximize in-domain performance to the same degree as a model optimized only for FF++.

5 Methodology

5.1 Model Architecture

The detector is built on a ViT-B/16 backbone [2], which is kept frozen during the main adaptation process. On top of this shared backbone, the model uses two complementary families of experts.

LoRA experts [4] are used to adapt global representations efficiently, enabling the detector to capture broad structural and semantic information without updating the full transformer backbone. In parallel, adapter experts are used to target local forgery cues. These include localized manipulations and high-frequency inconsistencies that may be overlooked by a purely global representation.

The overall architecture follows a Mixture-of-Experts design. A routing mechanism determines which experts should contribute most strongly for a given input, allowing the model to allocate computation adaptively across complementary feature pathways. Top- k routing, consistent with sparsely-gated MoE principles [3], is used so that only a subset of experts is emphasized for each sample, and load-balancing regularization is included to reduce expert collapse and encourage meaningful specialization across the expert pool. In this way, the model combines a stable transformer backbone with input-dependent expert adaptation for both global and local evidence.

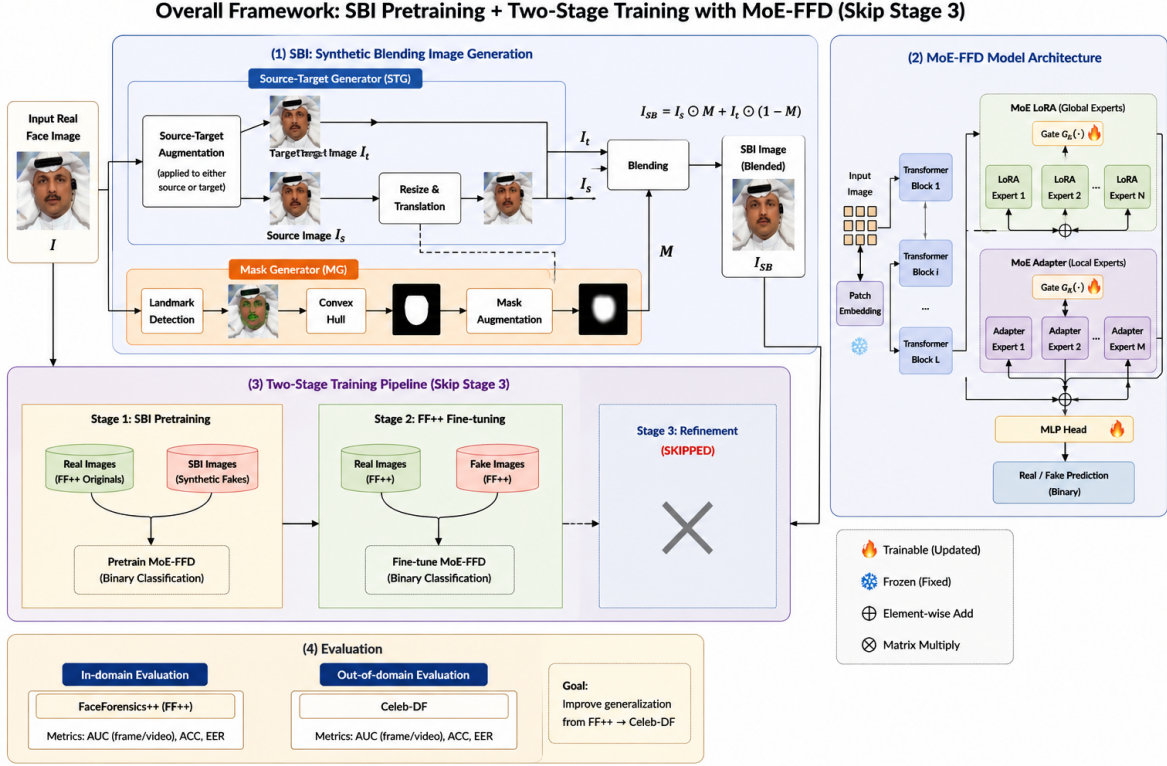


Figure 1: Architecture overview of the proposed MoE-based detector used in this project.

5.2 Training Pipeline

Two training settings are considered. The baseline setting trains the MoE-FFD detector directly on FF++ only for 10 epochs. This provides the reference condition for evaluating how well the architecture performs under standard single-dataset supervision.

The proposed training pipeline is a two-stage procedure. In Stage 1, the model is pretrained using SBI-style synthetic blended images [6] constructed from authentic FF++ images. This stage is intended to expose the detector to forgery-like inconsistencies before it encounters the downstream deepfake labels. In Stage 2, the pretrained model is fine-tuned on FF++ under the standard

supervised detection objective. No third-stage refinement is used in the final configuration reported here.

This staged design is motivated by the expectation that representation learning and task adaptation should play different roles. The first stage encourages broad forgery sensitivity, whereas the second stage aligns the model with the real benchmark distribution used for supervised training.

5.3 Datasets

FF++ is used as the in-domain dataset for training and in-domain evaluation. It provides the supervised benchmark on which both the baseline and proposed models are ultimately trained. Celeb-DF is used as the out-of-domain evaluation dataset to assess cross-dataset generalization.

These datasets are similar in task definition but different in distribution. They differ in their manipulation processes, artifact characteristics, visual quality, and overall data composition. This makes the FF++ to Celeb-DF transfer setting a useful test of whether the detector has learned general forgery cues or merely memorized benchmark-specific patterns.

5.4 Evaluation Metrics

Evaluation is conducted primarily with frame-level AUC and video-level AUC, which are standard measures for binary discrimination in deepfake detection. Where relevant, accuracy and equal error rate (EER) may also be reported as supporting metrics. The main emphasis, however, is on comparing in-domain and out-of-domain behavior in order to understand the trade-off between benchmark specialization and cross-dataset robustness.

5.5 Training Objective

The model is trained using a binary classification loss combined with a load-balancing regularization term for the Mixture-of-Experts (MoE) routing mechanism.

Specifically, the primary objective is a binary cross-entropy loss that encourages correct classification of real and fake samples. In addition, a load-balancing loss is applied to the MoE routing outputs to prevent expert collapse and promote balanced expert utilization.

The overall loss function can be expressed as:

$$L = L_{\text{classification}} + \lambda L_{\text{MoE}}$$

where $L_{\text{classification}}$ is the binary cross-entropy loss, L_{MoE} is the load-balancing regularization term, and λ is a weighting factor.



Figure 2: Representative examples from FF++, including one real sample and one sample for each manipulation type.



Figure 3: Representative examples from Celeb-DF, shown as two real-fake pairs for visual comparison.

6 Data Visualization and Examples

This section presents representative visual examples from the datasets used in this project. The purpose is to illustrate the visual characteristics of authentic and manipulated facial images and to highlight the differences in artifact patterns across datasets.

6.1 FaceForensics++ (FF++) Examples

Figure 2 shows one real image and one example for each manipulation type in FF++. The manipulated samples are arranged relative to the same identity in order to make visual comparison more direct.

6.2 Celeb-DF Examples

Figure 3 shows representative Celeb-DF samples arranged as two real-fake pairs, so that each manipulated image can be compared with a corresponding real image from the same identity.

FF++ contains multiple manipulation methods, each of which can exhibit distinct artifact patterns related to face synthesis, reenactment, or texture generation. By contrast, Celeb-DF generally

contains more visually realistic and subtle manipulations. This visual difference helps explain why models trained on FF++ may achieve strong in-domain performance yet degrade when evaluated on Celeb-DF under cross-dataset transfer.

7 Experiments

The experimental comparison considers two training configurations. The baseline model is the MoE-FFD detector trained on FF++ only for 10 epochs. The proposed method uses a two-stage training pipeline: Stage 1 performs SBI-style pretraining on synthetic blended images, and Stage 2 fine-tunes the pretrained model on FF++. No third-stage refinement is used in the final experiments.

FF++ is used for training and for in-domain evaluation, while Celeb-DF is used as the out-of-domain benchmark. This setup allows the comparison to measure not only benchmark-specific performance on the source dataset, but also transferability under domain shift. Evaluation focuses on frame-level AUC and video-level AUC, with accuracy used as a secondary metric when relevant. In the main comparison reported below, AUC is the primary measure because it captures discriminative performance consistently across datasets.

8 Results

Table 1 summarizes the main quantitative comparison between the FF++-only baseline and the best two-stage SBI-pretrained model. The proposed method improves the FF++ AUC from 0.9729 to 0.9770 and raises the Celeb-DF AUC from 0.7415 to 0.7647. As a result, the average AUC across the two datasets increases from 0.8572 to 0.8709.

The most important change is the improvement on Celeb-DF, where the proposed method gains 0.0232 AUC. This result indicates better out-of-domain generalization under the FF++ to Celeb-DF transfer setting. At the same time, the model also improves on FF++ by 0.0041 AUC, showing that the staged training pipeline does not merely trade in-domain performance for cross-dataset robustness in this configuration. Taken together, these results show a consistent overall improvement rather than a shift in performance from one dataset to the other.

The proposed method outperforms several prior methods on Celeb-DF, including Face X-ray, GFF, Xception, and F3Net. This comparison further supports the competitiveness of the staged SBI-pretrained MoE-based detector in the out-of-domain setting.

9 Discussion

The results suggest that SBI-style pretraining improves cross-dataset generalization by helping the model learn forgery cues that transfer more effectively beyond the source dataset. The gain on Celeb-DF is especially relevant in this regard, since Celeb-DF serves as the out-of-domain evaluation benchmark. A plausible interpretation is that Stage 1 exposes the detector to a broader range of

Table 1: Comparison between the FF++-only baseline and the best two-stage SBI-pretrained model.

Method	FF++ AUC	Celeb-DF AUC (CDF)	Average AUC
Face X-ray	-	74.20	-
GFF	-	75.31	-
ViT-B	-	72.35	-
Xception	-	64.14	-
RECCE	-	61.42	-
RECCE	-	69.25	-
SFDG	-	75.83	-
EN-B4	-	65.24	-
F3Net	-	68.69	-
Baseline (FF++ only)	0.9729	0.7415	0.8572
Two-stage SBI pretraining	0.9770	0.7647	0.8709

blended inconsistencies before task-specific fine-tuning, which encourages the learned representation to rely less on narrow FF++ artifacts.

An important aspect of the present result is that the best configuration improves both FF++ and Celeb-DF. Unlike settings in which cross-dataset gains come with a slight in-domain decline, the two-stage pipeline reported here yields stronger performance on the source and target datasets simultaneously. This pattern suggests better overall feature learning rather than a simple trade-off between specialization and robustness.

The staged structure of training helps explain this behavior. Stage 1 introduces general forgery patterns through SBI-style synthetic blending, providing a broader initialization for the detector. Stage 2 then adapts those features to the real deepfake distribution in FF++, allowing the model to retain general forgery sensitivity while aligning with the downstream task. In this sense, the two stages play complementary roles: the first emphasizes transferable manipulation cues, and the second calibrates those cues to the benchmark used for supervised learning.

A key implication of this finding is that data-centric pretraining can be effective even when the architecture itself is unchanged. The improvement is obtained without redesigning the MoE-FFD model, which suggests that the training pipeline is an important source of generalization gains. In the present setup, two-stage training is sufficient to produce a measurable benefit, and no additional refinement stage is required to obtain the strongest reported result.

These gains should nevertheless be interpreted with appropriate caution. The improvements are meaningful and consistent, but they are not large enough to justify broad claims about solving cross-dataset deepfake detection. Rather, the evidence supports a narrower conclusion: SBI-style pretraining is a practical way to improve the robustness of a strong MoE-based detector in this specific transfer setting. While the absolute performance differs from reported results in prior work due to the reproduction setting, the proposed method consistently improves over the baseline under the same experimental conditions.

10 Conclusion

This project addressed the problem of cross-dataset generalization in deepfake detection, focusing on transfer from FF++ to Celeb-DF. The proposed approach combined the MoE-FFD detector with SBI-style synthetic forgery pretraining in a staged training pipeline, while keeping the underlying architecture unchanged.

The experimental results show consistent improvements on both in-domain and out-of-domain evaluation. Relative to the FF++-only baseline, the two-stage SBI-pretrained model improves FF++ AUC, Celeb-DF AUC, and the overall average AUC. These gains indicate that the proposed approach improves generalization in a measurable and practically useful way.

Overall, the method is simple, data-centric, and easy to integrate into an existing training pipeline. The observed gains are moderate rather than dramatic, but they are consistent across the reported metrics. This makes the proposed approach a practical strategy for improving cross-dataset robustness without introducing a new model architecture or making overly strong performance claims.

11 References

References

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